Sahaj Bhadja

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Summary

Software engineer with 2+ years building portable applications, interactive features, distributed workflows, and automated testing using C++, Python, JavaScript, TypeScript, React, object-oriented design, and systematic debugging.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Built fault-tolerant FastAPI workflows processing 5,000+ monthly jobs with explicit states, validation, reconciliation, monitoring, and failure isolation, achieving 99%+ reliability across recurring production feature workflows.
- Architected Azure SQL and Blob Storage services for 500+ users, analyzing performance and tuning indexed access patterns to reduce interactive dashboard latency from 1.2 seconds to 350 milliseconds.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React interfaces and Flask APIs handling 10,000+ daily requests, translating grading requirements into responsive upload, review, history, feedback, and reporting features for 150+ university instructors.
- Designed concurrent processing for 1,000+ files using multithreading, secure ingestion, AWS RDS, and fault-tolerant execution, debugging complex failure paths while maintaining stable performance during peak workloads.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered portable microservices and 20+ automated PyTest and Mockito suites across Windows, Linux, and macOS, using Jenkins pipelines to validate nightly builds and produce actionable failure reports.
- Optimized REST API and ETL modules using object-oriented patterns, AWS Glue, S3, and SQL, improving response times by 30%, pipeline efficiency by 68%, and cross-platform systems maintainability.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a modular MERN application with REST APIs, WebSockets, and eight services supporting 2M+ synchronized records while translating defined requirements into responsive customer and inventory workflows reliably.
- Automated Docker-based builds, testing, and deployments through Jenkins CI/CD, reducing release time by 50% while improving integration consistency, defect resolution, and production delivery reliability for engineering teams.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built Flask APIs for a booking application supporting scheduling, payments, and status updates across 50,000+ records with synchronized Firebase-backed workflows, validation controls, and clear operational state tracking.
- Implemented webhook-based payment synchronization and root-cause troubleshooting, reducing manual handling by 70% while improving transaction visibility, error recovery, and responsiveness across distributed application components during production operations.

Skills

Programming: C++, Python, Java, C, JavaScript, TypeScript, SQL, Bash, object-oriented programming, data structures, algorithms, design patterns
Interactive Development: React, Next.js, HTML5, CSS3, responsive UI, state management, real-time updates, WebSockets, REST APIs, simulation, Gantt visualization
Testing & Quality: debugging, unit testing, integration testing, regression testing, PyTest, Mockito, test automation, defect isolation, code reviews, performance optimization
Platforms & Tools: Windows, Linux, macOS, Git, Docker, Jenkins, GitHub Actions, CI/CD, AWS, Azure, PostgreSQL, MongoDB, multithreading, distributed workflows, Agile/Scrum

Projects

CPU-Scheduling Simulator Live | *C++, React, JavaScript, EJ-2*
- Implemented six scheduling algorithms with interactive controls, dynamic metrics, and Gantt visualization across 1,000+ process combinations, reducing debugging time by 20% through deterministic simulation and edge-case testing.

PiXMod Live | *Next.js, TypeScript, CLIP, MongoDB, Transformers*
- Built an interactive visual application for segmentation, recoloring, restoration, and semantic search across 100,000+ indexed images, combining responsive interfaces with scalable metadata and model-driven processing workflows.

Chat2SQL Code | *Java, TypeScript, Docker, Maven, Node.js, Yarn*
- Built a portable database workspace supporting 10+ systems, using Docker, Maven, Node.js, and Yarn to standardize maintainable development and deployment workflows across Windows, macOS, and Linux environments.